

Université 8 Mai 1945 Guelma

Course : Business intelligence

4 ING - Artificial Intelligence

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Chapter 2 : Integration and Administration

What you will learn ?



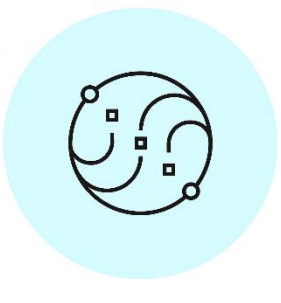
Describe what an ETL process is



Describe what data extraction means



Describe what data transformation means



Describe what data loading means



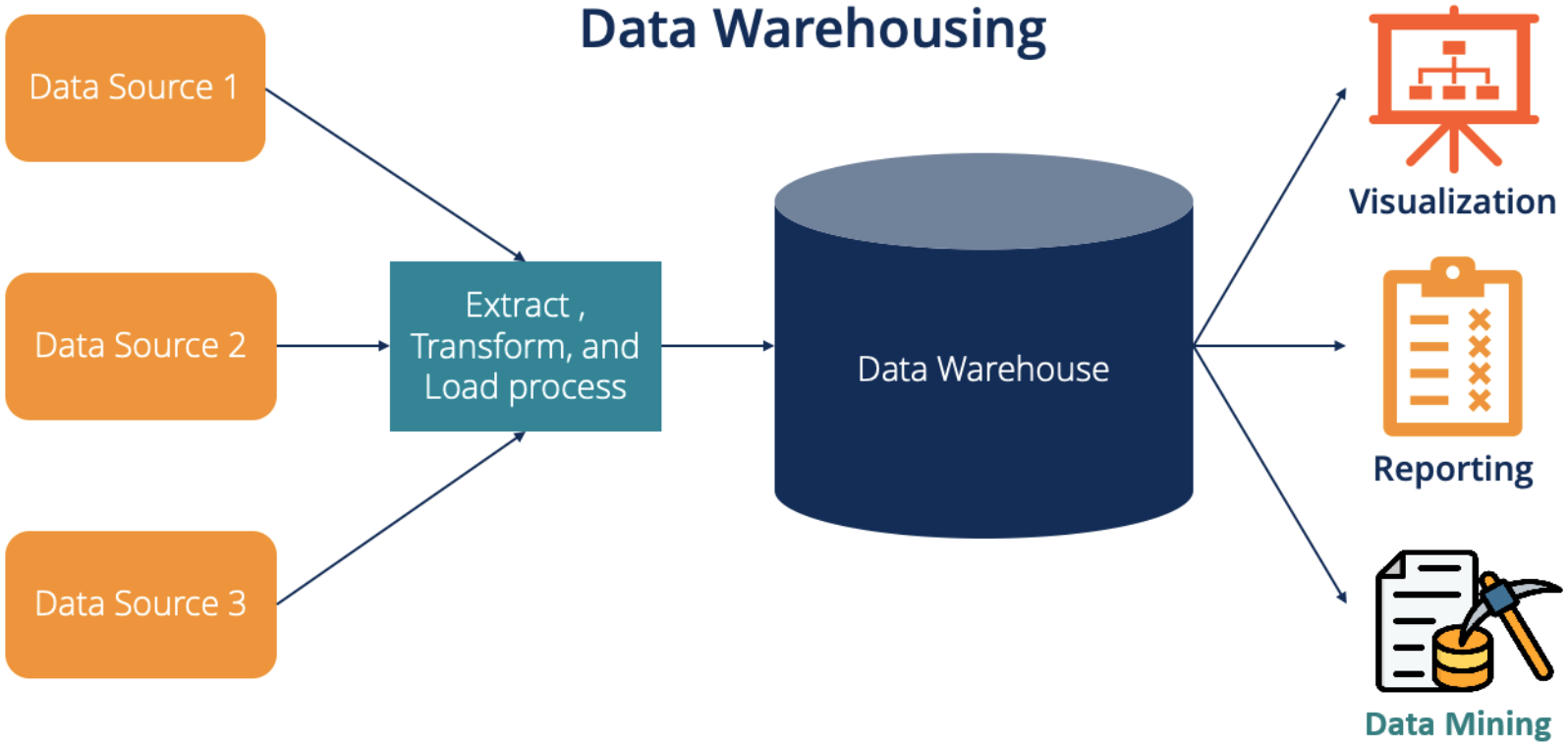
List use cases for ETL processes

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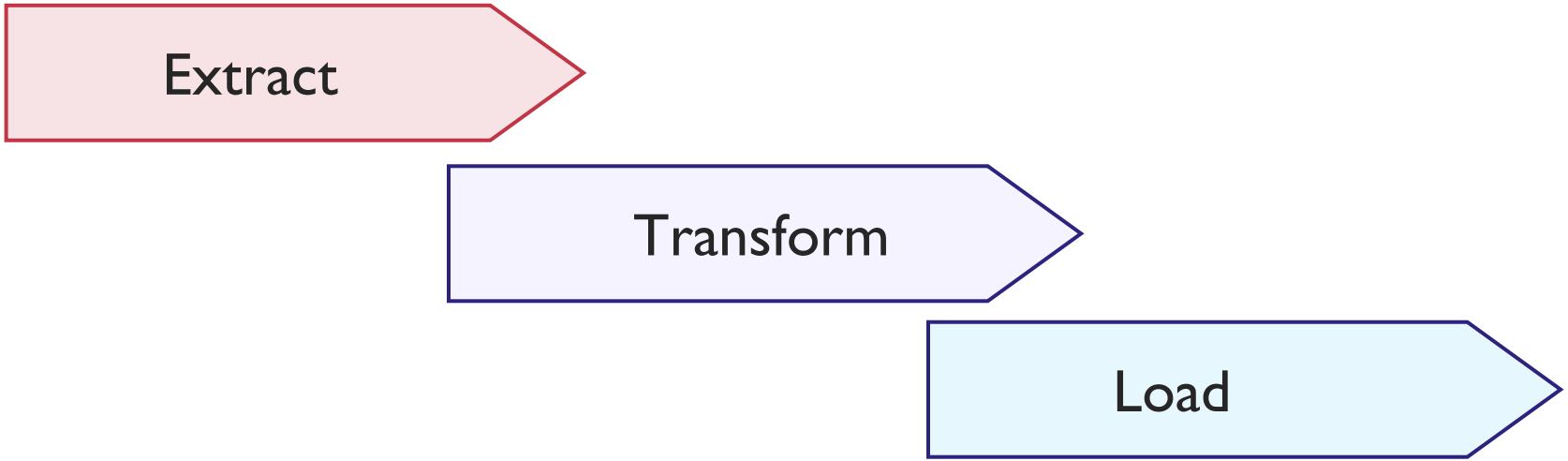
1.What is an ETL Process ?

2. Use cases for ETL pipelines
3. Data Extraction
4. Data Transformation
5. Data Loading

Recap: Data Warehouse Architecture

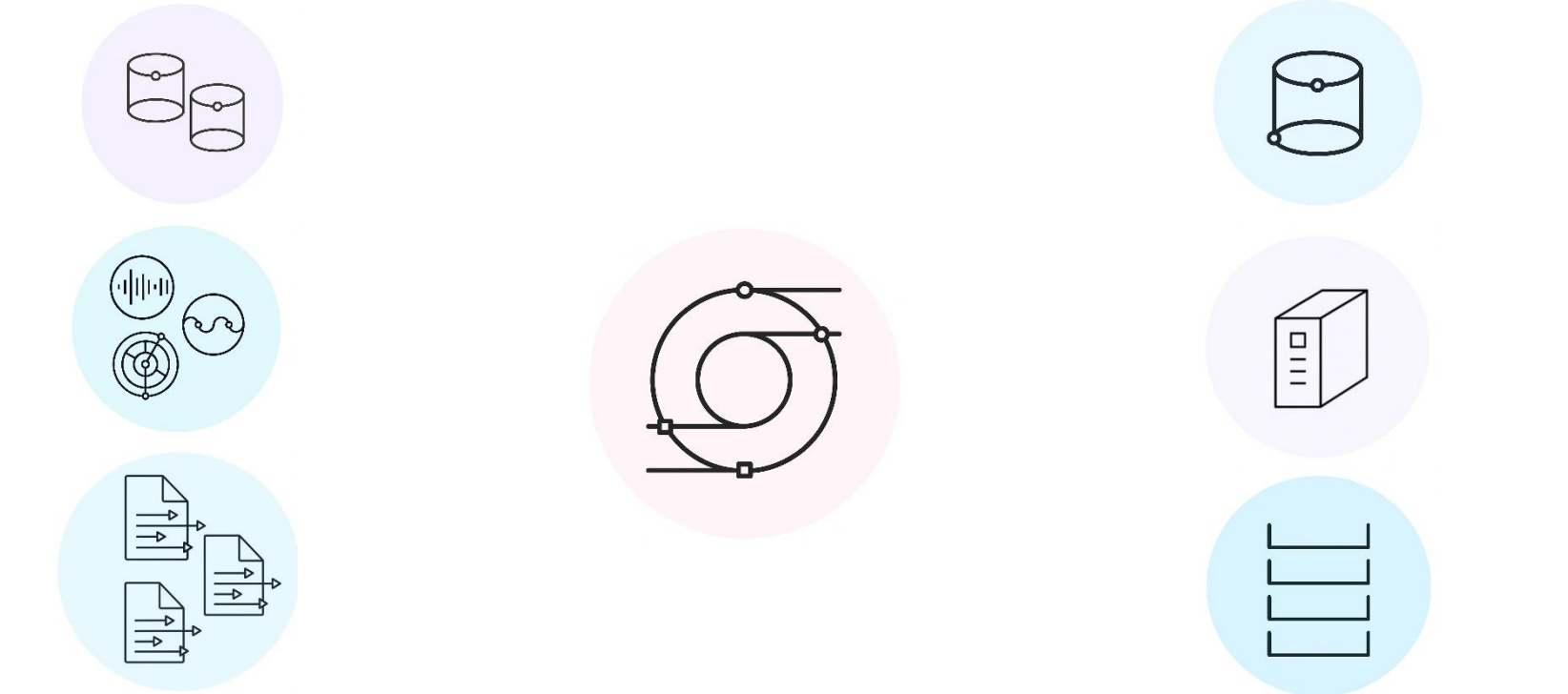


What is an ETL Process?



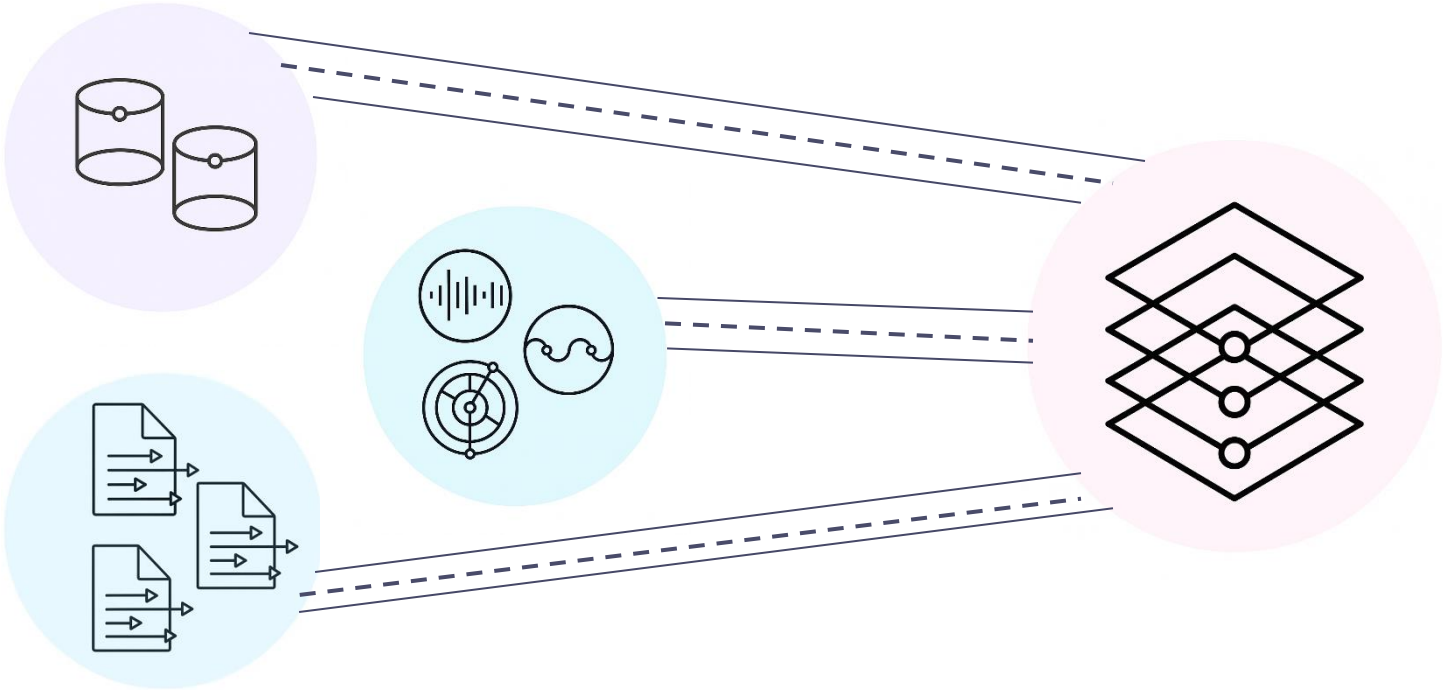
ETL is an automated process that collects data from different sources, cleans and organizes it, then loads it into a system like a data warehouse so it can be used for analysis.

What is an ETL Process?



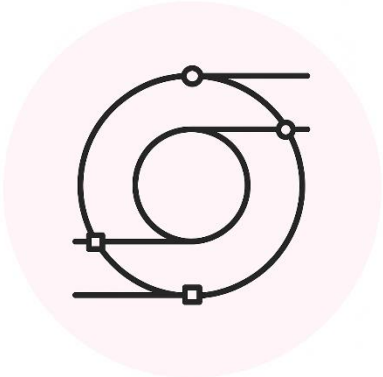
What is an ETL Process?

E = Extract: Extracting data from the source



What is an ETL Process?

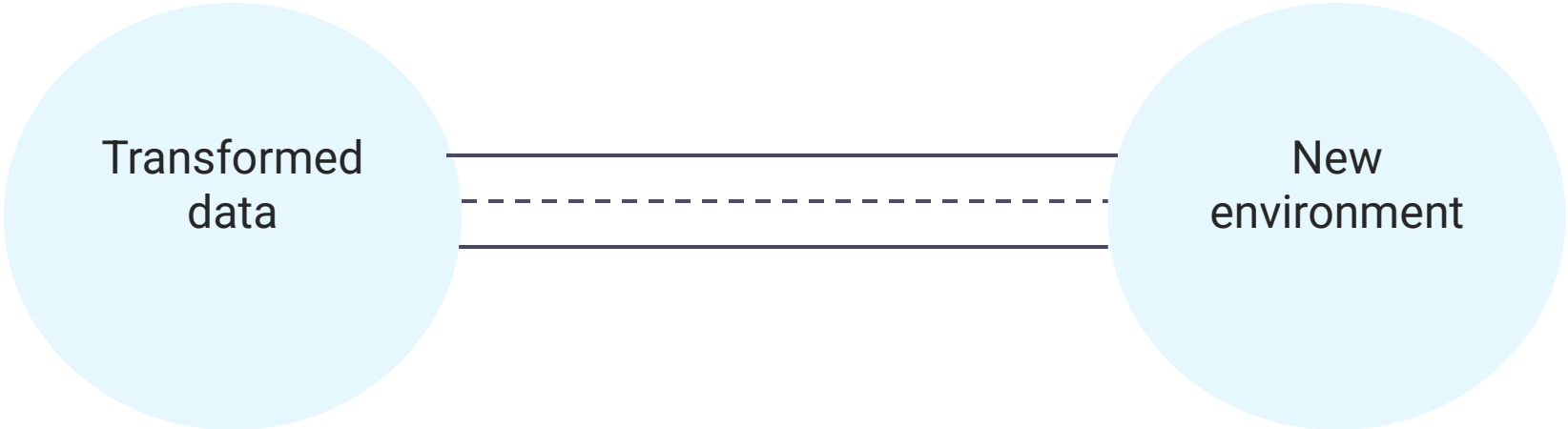
T = Transformation:



Transforming data into the format for the output

What is an ETL Process?

L = Load: Loading data into a data warehouse or other storage

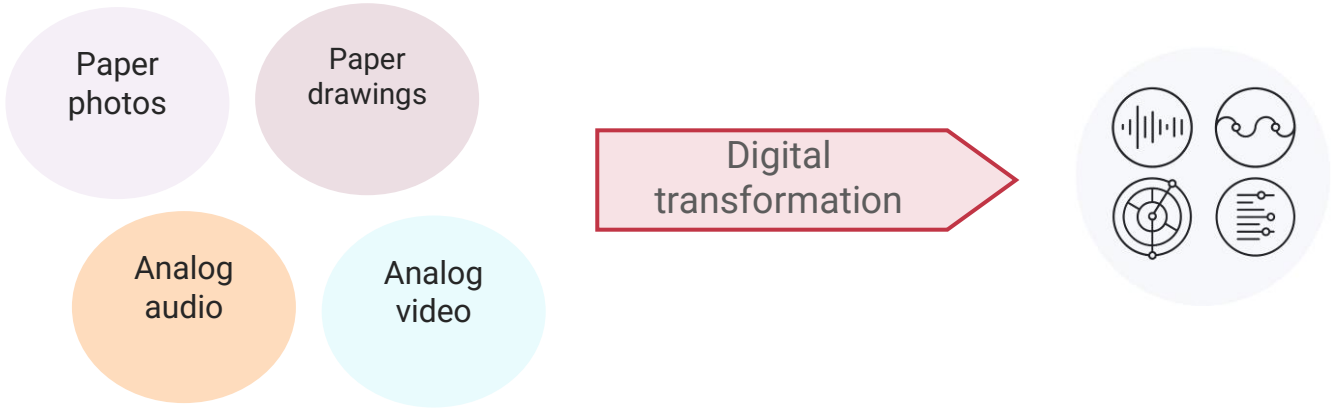


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1. What is an ETL Process ?
2. **Use cases for ETL pipelines**
3. Data Extraction
4. Data Transformation
5. Data Loading

Use cases for ETL pipelines

- Digitizing analog media



- **Dashboards:** use ETL pipelines to turn raw data into useful indicators (KPIs)
- **Machine learning:** Training and deploying ML models for prediction and augmented decision-making

Chapter 2 : Integration and Administration

1. What is an ETL Process ?
2. Use cases for ETL pipelines
3. **Data Extraction**
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What is extraction ?

- Extraction means accessing data and reading it into an application, usually through an automated process.

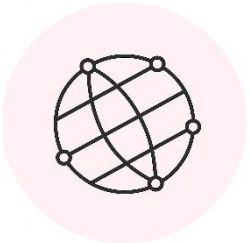
- It can be done by:
  - **Web scraping** – collecting data from web pages using tools like Python.
  - **APIs** – connecting automatically to data sources.

- The data may be:
  - **Static** (archives, updated in batches)
  - **Dynamic or streaming** (social media, IoT sensors, weather data).

Examples of raw data sources



Paper documents



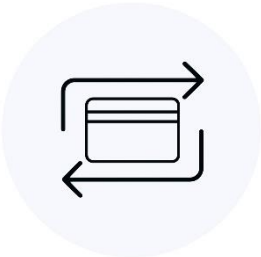
Web pages



Analog audio/video



Survey, statistics, economics



Transactional data

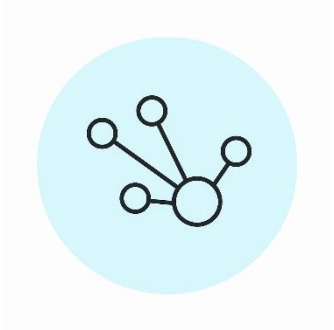
Examples of raw data sources



Social media



Weather station networks



IoT



Medical records

Techniques for extracting data

Data extraction techniques include:



OCR



ADC sampling, CCD sampling



Mail, phone, or in-person surveys and polls



Cookies, user logs

Techniques for extracting data

More techniques include:



Web scraping



APIs



Database querying



Edge computing

Use cases of data extraction

- Integrating disparate structured data sources via APIs
- Capturing events via APIs and recording them in history
- Data migration (direct to storage) for further processing
- Diagnosing health problems with medical devices

Chapter 2 : Integration and Administration

1. What is an ETL Process ?
2. Use cases for ETL pipelines
3. Data Extraction
4. **Data Transformation**
5. Data Loading

What is data transformation ?

Processing data so it fits the needs of the target system and the intended use case.

Key Transformation Processes:



Cleaning: Fixing errors or filling missing values.



Filtering: Selecting only the needed data.



Joining Sources: Merging related datasets.



Feature Engineering: Creating KPIs or new variables for analysis or machine learning.

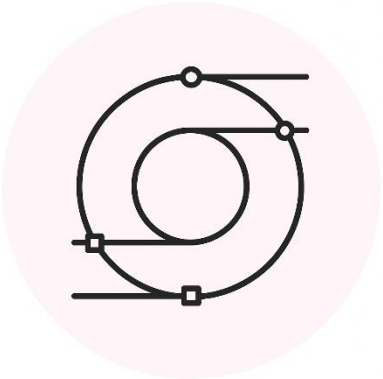


Formatting & Data Typing: Ensuring data is compatible with its destination.

Data transformation techniques

More techniques include:

- **Data typing**
- **Data Structuring**
- **Anonymizing, encrypting**

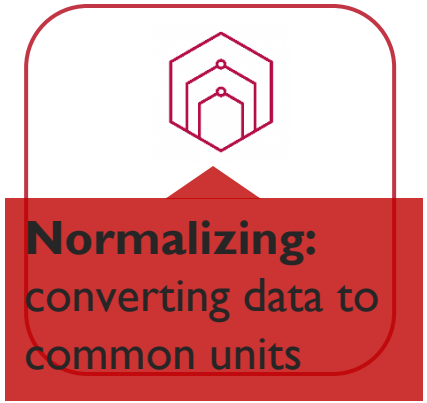


Data transformation techniques

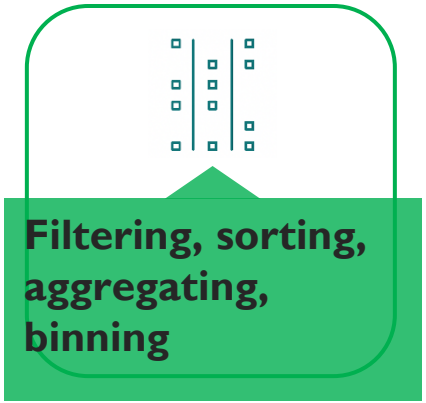
Other types of transformations include:



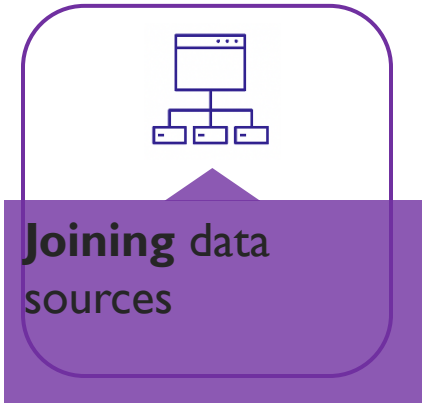
Cleaning: duplicate records, missing values



Normalizing: converting data to common units



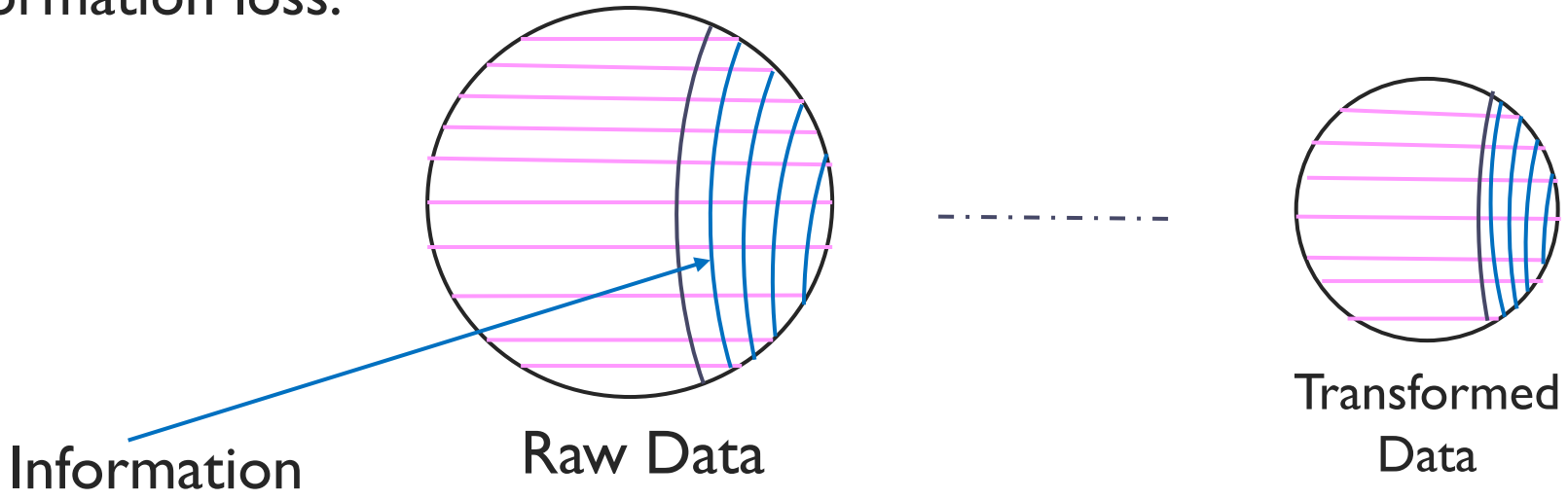
Filtering, sorting, aggregating, binning



Joining data sources

Information loss in transformation

Visualizing
information loss:



Information loss in transformation

Examples of ways information can be lost in transformation processes include:

- **Lossy Data Compression** – like converting floats to integers
- **Filtering** – permanently removing parts of data.
- **Aggregation –** summarizing data hides details (e.g., yearly vs daily).

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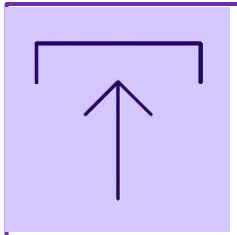
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5. **Data Loading**

What is data loading ?

- Data Loading means writing data to some new destination environment
- Typical destinations include databases, data warehouses, and data marts

Goal of data loading:
Making data readily available for analytics, dashboards, reports, and AI models

Data Loading strategies



Full Loading

Used for:

- Starting to track transactions in a **new data warehouse**
- Loading an **initial data history** into a database

This method is used when **no prior data exists** — it loads the entire dataset from scratch.



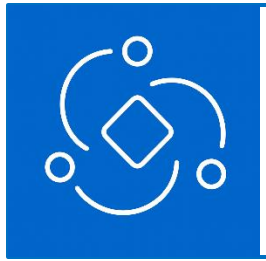
Incremental Loading

Used for:

- Loading **only data that has changed** since the last load
- **Appending new data** instead of overwriting existing records

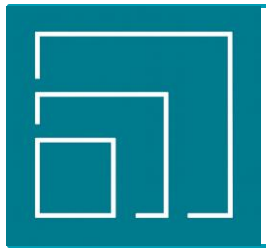
This strategy ensures **efficient updates** and supports **continuous data growth** after the initial full load.

Types of incremental loading



Stream Loading

- Data is loaded in real time



Batch Loading

- Data is loaded in batches

Stream Loading

- Stream loading refers to **continuous data updates** performed in the data warehouse or other storage systems **as new data arrives**
- **It is usually triggered by events:**
  - Real-time data, such as sensor data, social media feed, IoT devices
  - User requests, such as video or music streaming, web pages
  - …

Batch Loading

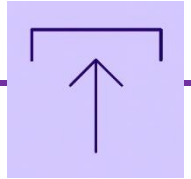
- Refers to **periodic updates** made/pushed to the data in the data warehouse or other storage systems, such as daily updates, hourly updates, or weekly updates.
- Batch data can be scheduled. For **example to run stock update every day at midnight**
- We can use:
  - Linux Cron jobs
  - Windows Task Scheduler



Push versus pull methodology

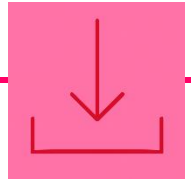
Push method

- Used when source pushes data to data warehouse
- Useful for real time data (stream loading)



Pull method

- Data warehouse pulls the data from the source
- Useful for scheduled extractions and batch loading



Loading plans

Serial or sequential loading

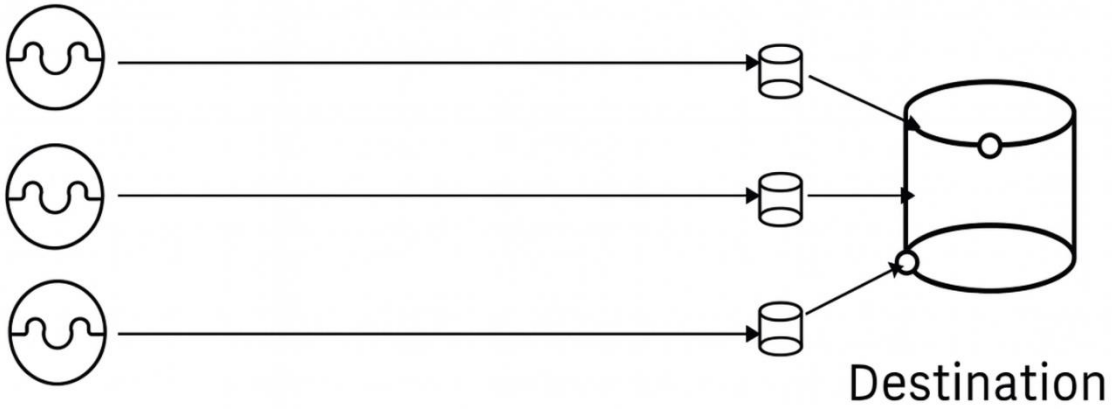
- Data is added one after the other in sequence
- This is how data loads in the data warehouse by default (Default plan)

Parallel loading

- Data from different sources are loaded parallelly
- Data from one source is split into chunks and loaded
- Faster and optimized approach

Parallel loading

Multiple Data Streams



Parallel loading

File Partitioning

